

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
				3-4 marks: At least five relevant points, discussing the oxidation states of two sets of elements <i>The candidate constructs a coherent account including most of the key elements of the indicative content and little irrelevant material, explaining at least one of the inert pair effect or octet expansion. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks: Two or three relevant points, discussing the oxidation states of any of the elements <i>The candidate attempts to link at least three relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material, however there is some understanding of the patterns in oxidation states. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks: The candidate does not make any attempt or give an answer worthy of credit.						
	(b)	(i)		rate = $k [\text{Pt}^{4+}] [\text{Sn}^{2+}]$	1		1			
		(ii)		(rate constant is large so) the reaction is very fast has a high rate (1) reaction would be complete / most would have reacted in 30 seconds (1)	2			1 1		1 1
	(c)	(i)		frequency that is absorbed by one reactant only but not by its product (or vice versa) accept 'frequency absorbed by only one species'	1		1			1